

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/228792208>

Fundamental mathematical structures applied to physics and biology.

Conference Paper · August 2005

CITATION

1

READS

127

2 authors:



[Peter Rowlands](#)

University of Liverpool

98 PUBLICATIONS 571 CITATIONS

[SEE PROFILE](#)



[Vanessa Hill](#)

19 PUBLICATIONS 235 CITATIONS

[SEE PROFILE](#)

Fundamental mathematical structures applied to physics and biology.

Peter Rowlands and Vanessa Hill

*Department of Physics, University of Liverpool, Oliver Lodge Laboratory,
Oxford St, Liverpool, L69 7ZE, UK. E.mail p.rowlands@liverpool.ac.uk
Department of Biological Sciences, Royal Holloway University of London,
Egham Road, Egham, Surrey, TW20 OEX, UK. E.mail
Vanessa.hill@rhul.ac.uk*

Abstract. Mathematical structures apparently underlying different aspects of physics and biology are examined in relation to their possible common origin in a universal system of process applicable to Nature.

The origins of the Dirac nilpotent

There are some tantalizing hints that a few simple mathematical structures play a significant role in both physics and biology, when taken at the fundamental level. A key connecting idea is 3-dimensionality. Using the concept of a universal computer rewrite system and starting from the idea that zero totality is necessary at all times [Rowlands and Diaz, 2002, Diaz and Rowlands, 2004], we can devise an algebraic structure in which 3-dimensionality, through the linked property of anticommutativity, generates the entire system of discrete numbering on which mathematics is founded. At the same time, replicating biological systems seem to be generated in a way that conforms to similar mathematical processes, and to be constrained by the necessity of fitting into a context determined by the 3-dimensionality of space. It is, of course, one thing to hypothesize that this physics / biology link exists, another to prove it; but the link, if it exists, would be so significant that is worth examining a number of possible connections suggested by the underlying mathematics.

A significant question in the creation of all systems with large-scale order is how complexity can arise from simplicity. Related to this is the question of how asymmetry can arise from symmetry. The universal rewrite system suggests how this can occur in physics, and it will be interesting to see if the same applies in biology. In the rewrite system, anticommutativity is seen as the only true source of discreteness in nature. If **a** and **b** are anticommutative with each other, then neither can be anticommutative with anything else, except **ab**. So **a**, **b** and **ab**, or **i**, **j** and **ij**, form a closed or discrete set. The rewrite process generates an infinite succession of otherwise identical closed sets of this kind, which

thereby generates a system of numbering. It also generates a series of Clifford-type algebras:

real scalar	units ± 1
complex (real + imaginary) scalar	units $\pm 1, \pm i$
quaternion	units $\pm i, \pm j, \pm k$
multivariate vector (complex quaternion)	units $\pm i, \pm j, \pm k$, etc.

Quaternion and imaginary scalar units are, of course, square roots of -1 . Multivariate vector and real scalar units are square roots of 1.

The total number of algebraic units (the group structure) doubles at each successive stage, and, in principle, continues doubling to infinity. Here the successive orders are 2, 4, 8, 16, and the doubling mechanism requires one of three possible operations: conjugation (applying $-$ as well as $+$); complexification (multiplying by an additional complex factor, so 1 becomes $1, i$ and i, j, k becomes $ii = i, ij = j, ik = k$); and dimensionalization (in which a complex factor, such as i is joined by its anticommutative partner j). Here the successive operations are:

order 2	conjugation
order 4	complexification
order 8	dimensionalization
order 16	complexification, etc.

Conjugation occurs only at order 2. After this there are successive occurrences of complexification and complexification. So the cycle repeats at order 16. To *combine* the separate orders at which we obtain conjugation, complexification, dimensionalization and repetition (i.e. the first four algebras) requires order 64.

Remarkably, the algebras required for these four orders are respectively those required for the four fundamental physical parameters: mass (real scalar), time (pseudoscalar = imaginary scalar), charge (quaternion) and space (multivariate vector).

time	space	mass	charge
pseudoscalar	vector	scalar	quaternion
i	$i \ j \ k$	1	$i \ j \ k$

The combination of these requires an algebra of 64 units (including $+$ and $-$ signs). This is made up of

4 complex scalars	$(\pm 1, \pm i)$
12 complex vectors	$(\pm 1, \pm i) \times (i, j, k)$
12 complex quaternions	$(\pm 1, \pm i) \times (i, j, k)$
36 complex vector quaternions	$(\pm 1, \pm i) \times (i, j, k) \times (i, j, k)$

This is, of course, also known as the Dirac algebra because it is the algebra which occurs the Dirac equation for the electron, although there it is often, more conventionally, expressed in terms of the five γ matrices. The significant thing here is that the algebra does not need 8 primitive

units, and + and – signs, to generate these 64 parts. *It needs only 5 composite ones.* In effect, the most efficient structure is not the most primitive one, and is also not the most symmetrical. The 1 + 3 / 1 + 3 symmetry that can be observed in the 8 primitive units is completely broken, when we write down the 5 composite units that most efficiently produce the 64-component algebra, because this can only be done by taking one of the 3-dimensional quantities (space or charge) and superimposing its units on the others. If we take charge (the conserved quantity), we obtain:

$$ik \quad ii \quad ji \quad ki \quad lj$$

These five combined or composite units also acquire a composite physical character, derived from their components, respectively, energy (E), three components of momentum ($\mathbf{p}$), and rest mass (m):

$$\begin{array}{ccccccc} E & ip_x & jp_y & kp_z & m \\ & = \mathbf{p} & \text{(when components combined)} \end{array}$$

That is, the charges produce new quantities that are quantized and conserved (like themselves) but also retain the respective pseudoscalar, vector and scalar aspects of time, space and mass. In complete form, we have:

$$\begin{array}{ccccccc} ikE & iip_x & jip_y & kip_z & ljm \\ ikE & ip & & & ljm \end{array}$$

A system made of these components is the most efficient packaging of the information concerning the four fundamental parameters, space, time, mass and charge that can exist in nature. In fact, no other type of physical system exists. We describe it as a fermion (or antifermion, depending on the sign of ikE). It turns out that the fermion / antifermion described by the combination ($\pm ikE \pm ip + ljm$) is the fundamental unit of physics. It packages the space, time, mass and charge components into a single unit. Very significantly, every packaging of this kind is a nilpotent or *square root of zero*. This means that it squares to zero with the ‘vacuum state’ – ($\pm ikE \pm ip + ljm$) created simultaneously when we extract it from a zero totality superposition. Another way of expressing this (which is derivable from certain mathematical properties of the nilpotent) is to say that it has a ‘spin’ describable in half-integer values, which becomes ‘single valued’ (full integer) when combined with its vacuum state or equivalent. (NB No new information would be gained by allowing a sign variation in lm in a fermion state, as opposed to vacuum.)

While the symmetry-breaking creation of the concepts of energy (E), momentum ($\mathbf{p}$) and rest mass (m) is the immediate result of the fermionic packaging, the process also works simultaneously in the opposite direction by breaking the symmetry between the three charges (i, j, k) as

they take on the respective pseudoscalar, vector and scalar aspects of time, space and mass. So we now have, for charge:

$$\begin{array}{ccc}
 \text{weak} & \text{strong} & \text{electric} \\
 w & s_R \ s_G \ s_B & e \\
 & (3 \text{ colours}) & \\
 i\mathbf{k} & i\mathbf{i} \ j\mathbf{i} \ k\mathbf{i} & 1j \\
 \text{pseudoscalar} & \text{vector} & \text{scalar}
 \end{array}$$

The significance of the pseudoscalar term

Two very significant facts are associated with defining $(\pm i\mathbf{k}E \pm i\mathbf{p} + 1j\mathbf{m})$ as a nilpotent. The first is that, to ensure nilpotency, one term at least must be pseudoscalar, that is, contain the factor i . Otherwise the squared terms will not add up to zero. The pseudoscalar is the term that becomes the ‘book-keeper’ (energy, weak charge), the creator and destroyer, the aggregator and disperser of matter in all its forms. In the universal rewrite system, this factor occurs as part of an incomplete quaternion set – there are no intrinsic pseudoscalars – and it is in removing this incompleteness that all physical interactions are ultimately situated. Here, the pseudoscalar i occurs in the position occupied by the parameters time, energy and weak charge, and the incompleteness manifests itself in the properties of all these quantities. The second significant fact is that a pseudoscalar quantity is, by definition, mathematically ambiguous. We cannot distinguish between $+$ and $-$ signs, *but both must be present*. This is what allows us to generate antifermions $(-E)$ or $(-w)$ and time reversal symmetry $(-t)$. The conditions are, of course, connected: antifermions do not have negative energy in ordinary time; they only have negative energy in reversed time. The result of the sign ambiguity in the pseudoscalar term is that the fermion and vacuum are always a dual combination.

Though the Dirac nilpotent is structured as $(\pm i\mathbf{k}E \pm i\mathbf{p} + j\mathbf{m})$, it is also often convenient to multiply throughout by i and write it as $(\pm \mathbf{k}E \pm i\mathbf{ip} + ijm)$. The bracketed expressions are really column (or row) vectors whose four components take up the four sign variations of $\pm \mathbf{k}E \pm i\mathbf{ip}$. The four components represent the fermion state itself and three vacuum ‘reflections’ representing the effects of the three charges. In other words, the nilpotent, when structured in the column vector form $(\pm \mathbf{k}E \pm i\mathbf{ip} + ijm)$, incorporates both energy and charge information. Information about charge ‘occupancy’ is determined by relative phases. The three vacuum ‘reflections’ are structured as $\mathbf{k}(\pm \mathbf{k}E \pm i\mathbf{ip} + ijm)$, $\mathbf{i}(\pm \mathbf{k}E \pm i\mathbf{ip} + ijm)$, $\mathbf{j}(\pm \mathbf{k}E \pm i\mathbf{ip} + ijm)$, which represent respective the weak, strong and electric vacua. If we multiply $(\pm \mathbf{k}E \pm i\mathbf{ip} + ijm)$ by any of these any

number of times it has no effect. We can consider the three things as ‘partitions’ of a total, or gravitational, vacuum, which would be $1 (\pm \mathbf{kE} \pm i\mathbf{ip} + i\mathbf{jm})$. Multiplying $(\pm \mathbf{kE} \pm i\mathbf{ip} + i\mathbf{jm})$ by this would make it disappear (Pauli exclusion). (NB The reason why there is no independent m sign change is that there are only three charges. This has very important consequences in ensuring that there are only positive values for mass and ultimately energy, as also for proper time.)

The whole meaning of nilpotency is that the object squares itself to zero, and, from the rewrite mechanism which generates it, it would appear that the reason is that a fermion can have no existence outside of the total vacuum or ‘rest of the universe’. In this sense, the total vacuum is a continuum (not partitioned), whereas the three ‘partitions’ are discrete. We create a fermion only with its other half. The total effect remains zero. A fermion always acts in such a way as it is trying to find this other half. However, their only perfect partner is the completely delocalised vacuum (or rest of the universe). Fermions can be pictured as incomplete systems which are forever seeking the ideal partners needed to reduce them immediately to zero. This is the reason why we have aggregated matter. We can consider this by analogy with Newton’s third law of motion. Body A doesn’t act on body B but on the rest of the universe, and the same applies for body B supposedly acting on body A. However, body B can become, *in effect*, the rest of the universe if it is, say, very close to body A. (Thermodynamics is relevant here, and the distinction between closed and open systems; the nilpotent fermionic state is unique in being an open system, but defined to conserve energy.)

In the same way, we can create local aggregates of fermionic matter by a fermion, say $(\pm \mathbf{kE}_1 \pm i\mathbf{ip}_1 + i\mathbf{jm}_1)$, effectively finding its other half in another fermion, say $(\pm \mathbf{kE}_2 \pm i\mathbf{ip}_2 + i\mathbf{jm}_2)$. (Of course, ‘fermion’ is a generic term here: either or both of these states could be antifermionic.) The equivalent of the squaring operation here is the ‘vertex’ $(\pm \mathbf{kE}_1 \pm i\mathbf{ip}_1 + i\mathbf{jm}_1) (\pm \mathbf{kE}_2 \pm i\mathbf{ip}_2 + i\mathbf{jm}_2)$, which eliminates all quaternionic components when all four product terms are combined; and, if we make the specification that all products of fermions are non-quaternionic, then we only need to specify the lead term in an expression such as $(\pm \mathbf{kE}_1 \pm i\mathbf{ip}_1 + i\mathbf{jm}_1)$, say $(\mathbf{kE}_1 + i\mathbf{ip}_1 + i\mathbf{jm}_1)$; the other three terms become automatic.

An interaction occurs where the creation of a vertex leads to changes in the E and / or $\mathbf{p}$ terms of the components; and where the E and / or $\mathbf{p}$ become equalized, the vertex becomes a new combined, bosonic, state, with a purely scalar value. Such a state can only exist at all, however, if the two halves have different signs for either $\mathbf{kE}$ or $i\mathbf{ip}$, or both, which the combining force (the weak interaction) won’t recognize. The three

possibilities create spin 1 bosons (reverse E in the second fermion), spin 0 bosons (reverse E and $\mathbf{p}$ in the second fermion), and Bose-Einstein condensates (reverse $\mathbf{p}$ in the second fermion). Bose-Einstein condensation is a particular case of a very general physical concept called the Berry phase, with many distinct manifestations – other examples are the Aharonov-Bohm effect, the quantum Hall effect and the Jahn-Teller effect. Fermions, like electrons, self-aggregate, for example, with something like a nucleus or a magnetic field line to create a system with single-valued spin (not multiples of $\frac{1}{2}$). It is probable that Berry phase is manifested in some way with all fermions – we just don't see it in most cases, because the system with which it is connected in this way is too dispersed. It is through this kind of phenomenon that matter self-aggregates over long distances.

Ultimately, quantum mechanics is about relating the packaged and conserved energy and momentum terms in the nilpotent expression $(\mathbf{k}E + i\mathbf{p} + ijm)$ to the nonconserved parameters time and space, from which they were originally derived; and to do this we make $(\mathbf{k}E + i\mathbf{p} + ijm)$ into a differential operator in which E and $\mathbf{p}$ represent the quantum operators $\partial / \partial t$ and ∇ , expressing the variation on time and space. In the case of the system not being free or isolated (which is, of course, always true in reality), we also add on respective scalar and vector potential energy terms to these operators, which reflect the particular fields (i.e. other fermionic states) to which the fermion is subject. (Again, in reality, this will be an infinite number, though in practice one or a few may be dominant.) What happens now is that the operator requires a *unique* phase term on which to operate (or unique within the symmetry constraints of the fields involved). The result of the operation will then be an amplitude, which will always square to zero. So, in the case of a free particle, the operator $(\mathbf{k}\partial / \partial t + i\mathbf{p} + ijm)$ generates the uniquely defined phase term $\exp -i(Et - \mathbf{p}\cdot\mathbf{r})$, which, when differentiated by the operator, produces $(\mathbf{k}E + i\mathbf{p} + ijm)$, which squares to zero.

In more complicated cases, where the operator E is not simply $\partial / \partial t$, we get a completely different phase. A standard example is the hydrogen atom, where the inverse quadratic force (Coulomb) requires $E \rightarrow \partial / \partial t + A / r$, and the phase term has a real, not imaginary, exponential. For a discrete point particle, with spherical symmetry in space, it can be shown that there will be no solution unless there is a Coulomb term in E . We can also show that there are only three solutions with spherical symmetry: inverse quadratic force (Coulomb); inverse quadratic plus constant force; and inverse quadratic plus anything else. The first gives the characteristic electric force solution; the second produces confinement, with the characteristics of the strong interaction; the third is a harmonic oscillator,

irrespective of the actual nature of the force law. This is a creator and destroyer, with the characteristics of the weak interaction and the van der Waals forces responsible for aggregated matter. It is exactly what we would expect for dipolar and multipolar forces of any kind.

Ultimately, $(kE + i\mathbf{p} + ij\mathbf{m})$, in whatever form it is, uniquely determines the phase to be associated with it, and also the amplitude. The phase can be seen as corresponding in many ways to what we call ‘vacuum’, while the amplitude corresponds to the particle. In a sense, the phase is the connection of the particle to the rest of the universe, the carrier of nonlocality, the origin of the idea of phase conjugation (which determines locality) and the holographic principle. It is like a signal sent out by a fermion asking other fermions, etc., to organize themselves with respect to it, and to aggregate. It is also a fundamental part of the intrinsic duality of the nilpotent fermion. We have amplitude on the one hand, and operator plus phase on the other, and they must be identical.

Spin and aggregation of matter

The mechanism which results in the creation or destruction of (combined) bosonic from or into (uncombined) fermionic states is described as the harmonic oscillator. The harmonic oscillator is a classic indication of aggregation or complexity in a system. It is a statement that no system is ever ‘closed’. All fermions interact with each other; E and $\mathbf{p}$ are never fixed. This lack of closure is an expression of the second law of thermodynamics, and is the driver for all processes. No process is reversible. Time is unidirectional.

To arrive at the standard quantum mechanical harmonic oscillator, means realising that spin $\frac{1}{2}$ is intimately connected with the weak interaction, and results from a particular aspect of it (its incompleteness). Spin $\frac{1}{2}$ in physics comes from a fermion only being created simultaneously with its vacuum, and the spin is defined with respect to vacuum. A fermion cannot be defined otherwise. As we have already stated, it is like an interaction in the case of Newton’s third law – not body A on body B, but body A on the rest of the universe (which is mostly B), and body B on the rest of the universe (which is mostly body A). Vacuum is the rest of the universe. It is a manifestation of the nonlocal aspect of quantum mechanics (and is probably the same thing as ‘gravity’ – again relating to Newton’s third law). Now we can imagine the spin of the whole system as a kind of ‘helical’ motion. Interestingly, it is *double helical*, because the vacuum ‘spins’ $\frac{1}{2}$ at the same time as the fermion. The total is spin 1 (or 0), so the combination produces an ordinary 2π rotation, though each half rotates through 4π .

Another interesting aspect of spin is that it is generated from the vacuum via the weak interaction. The weak interaction is dipolar (a dual $\pm$ source), because of the sign ambiguity in the pseudoscalar parent quantity, and the basic force law for dipole-dipole is inverse quartic ($\propto r^{-4}$), just as the basic unipole-unipole law is Coulomb or inverse quadratic ($\propto r^{-2}$). In physics, *any* third body (or ‘pole’) introduced into a system where two bodies (or ‘poles’) have a $1 / r^2$ attraction / repulsion, will produce an additional $1 / r^4$ dependence. In other words, the $1 / r^4$ dependence is a natural result / expression of aggregation.

The most famous example of inverse quadratic plus inverse quartic in physics is the analogous perihelion precession of the planets produced by gravity. That is, while the inverse quadratic term produces the orbit, the inverse quartic term makes the orbit spin. So, we can consider the weak term as generating the spin. (In effect, the $\frac{1}{2}$ -integral ‘spin’ can be seen as the weak interaction operating a continual switching between the $+E$ and $-E$ or $+t$ and $-t$ states (*zitterbewegung*), or fermion and vacuum.) In the planetary case, the main reason for the inverse quartic term is the disturbing effect of other planets. That is, the orbit of Mercury round the Sun is disturbed by all the other planets in the solar system. Another way of expressing this is to say that the tendency to aggregation within the Sun-Mercury combination is supplemented by the tendency to aggregation between this combination and other aggregated bodies of matter. Any tendency to such aggregation creates a dipolarity.

The weak term is indeed the crucial one for the collective behaviour of matter. Matter becomes collective only when it overcomes the weak Pauli exclusion (the uniqueness of each fermionic state) by creating *physical* dipoles, rather than matter-vacuum dipoles. (This can be seen as a kind of localization.) It has been previously suggested that spin is a manifestation of a weak dipole moment [Rowlands, 2003b]. That is, it shows a bias to one sign over the other. This does not happen with strong and electric interactions. Particle physicists attribute the left-handedness of the weak interaction for fermions (and right-handedness for antifermions) to the idea that there is a filled weak vacuum. It is notable that the weak charge and energy are located in the same place in the Dirac nilpotent. (We note here also that collective matter has a left-handed bias in the same way as the weak interaction: L-amino acids dominate over D-amino acids, and it is tempting to conclude that this must have the same origin.)

Continuous vacuum energy, such as we would expect from a ‘filled’ vacuum, is what we mean by nonlocality. It is the continuing connectedness, through the vacuum, of apparently discrete fermionic states, and it is required to maintain Pauli exclusion. Rest mass is a

localization and therefore discretization of the continuous total vacuum energy. The continuity of vacuum energy is the reason for the left-handed bias of fermionic states, but, in discretizing it as rest mass, we also allow an element of right-handedness to emerge (as also E and $\mathbf{p}$).

Self-organization of matter

All processes in the entire universe can be considered as corresponding to the elimination of those aspects of the structure of the fermion which keep it separate from the rest of the universe: weak charge; $\frac{1}{2}$ -integral spin; a single energy state. In the first instance, fermions aggregate to become bosons or boson-like states, but this then proceeds to higher levels. In general, physical systems self-aggregate through the dipole-dipole van der Waals force, a classic expression of the action of vacuum, and, as we have seen, the concept of aggregation – in effect, complexity – originates, in physics, in the harmonic oscillator. Physically, the ‘oscillator’ aspect is seen in the behaviour of the molecules responsible for the various physical states of matter – gases, liquids and solids – which also represent different manifestations of the van der Waals force.

We think of this force as being electric or electromagnetic, because the components of matter are electrically charged, and so there are significant electrical forces involved; but the reason for the aggregation in the first place concerns the weak force. Any tendency for matter to aggregate is all about overcoming the weak force of Pauli exclusion or $\frac{1}{2}$ -valued spin. Any spin $\frac{1}{2}$ object, or any object with spin $\frac{1}{2}$ components, has a tendency to try to effect a physical realization of the ‘rest of the universe’. In terms of weak charges, this is like trying to cancel them. So $+w$ cancels $-w$, or, since the sign of the weak charge is ambiguous, $+w$ cancels $+w$. But this is only a tendency – it can never be satisfied, because a ‘real’ partner can never cancel out a state completely.

Hydrogen bonding is one of the classic dipole-dipole van der Waals forces, and, of course, it is precisely this which keeps the bases together in the two strands of DNA. The strands of DNA can be thought of as like the fermion and its partner (and they are equally subject to harmonic oscillation, as the links continually break and reform). All phase transitions necessarily involve the van der Waals force, because what is happening in a phase transition is that seemingly ‘independent’ systems are being more closely connected, apparently isolated systems are being realised as being connected with other apparently isolated systems. Anything which disturbs an apparently isolated (canonical, energy-

conserving system) manifests itself as dipolarity, leading to a van der Waals force, and a physical manifestation of helicity.

The formation of a nucleus is a classic phase transition. Even though the nucleus is held together by a van der Waals-type version of the strong force (a multipolar remanant of the quark-quark forces within the individual protons and neutrons), the real thing which makes it possible is the cancelling out of the fermionic nature of its components, or their weak charges. A phase transition occurs when there is a significant change in the number of independent states of energy and momentum in a system. Ultimately, phase transitions which decrease the number of independent states of energy and momentum (e.g. the formation of real bosons by the equalization of energies of fermion and antifermion) increase the amount of order in the system and decrease the entropy (and complexity); those which increase the number of independent states of energy and momentum decrease the order and increase the entropy. Natural processes will always favour the latter because the tendency is to multiply connections. We might suppose also that the emergence of higher structures at this stage follows a fundamental mathematical pattern guaranteeing maximum efficiency. A Bose-Einstein condensation is another classic example of a phase transition. Here the number of independent states of energy and momentum is decreased by the fact that many 'bosonic' states can be aggregated into a single state of energy and momentum because bosonic wavefunctions are scalar and so are not subject to Pauli exclusion.

DNA and RNA appear to behave in a way pre-determined by the mathematical structure required for nilpotency, and for a totality which always remains zero. The nilpotent fermion replicates itself in the vacuum, effectively through its phase, and it may be that a similar thing is involved in DNA. There is no reason to believe that the chemistry is unique (and, of course, A in DNA is replaced by U in RNA), but it may well be the case that, as soon as nature hit on a mode of replication, it took off and replicated. So aggregations happened until a particular replicating one developed. A universal phase effect must certainly have been involved.

The filled weak vacuum and the one-handed bias in nature

It is important to recognize that the force responsible for the emergence of higher-level structures in Nature is a strictly nonlocal or vacuum one. In fact, the organizing factor in all aggregation of matter is the weak force between fermion and vacuum. The fermion effectively has an unpaired weak charge which can always interact weakly with vacuum,

because the weak vacuum is *filled*. In principle, this is precisely what weak filled vacuum means: there is always something with which the fermion interacts, and it is nonlocal. It is the filled vacuum that is responsible for spin $\frac{1}{2}$, chirality, *zitterbewegung* (vacuum fluctuation), van der Waals dipolarity and rest mass. Its manifestations very likely include the prevalence of matter over antimatter, thermodynamics, and the arrow of time, and its origin comes from the fact that mass-energy and time are both continuous concepts, even in the discrete fermionic state.

As we have seen, the fermion's whole objective is to cease to exist, but this cannot happen because it would require the delocalised whole of the rest of the universe to become localised at the same point. However, the thing that makes it want to do this – the unpaired weak charge – can do this by combining with a fermionic / antifermionic partner in the bosonic state. However, many bosonic states are not a true union, and the structure remains open; for example, in the hydrogen atom – the parts are still distinct and try to combine with other fermions / bosons – e.g. the H atom becomes H₂ molecule, etc. The van der Waals force which drives the process originates in the fermion's weak interaction with vacuum. Though the ordinary weak interaction between two real particles or weak charges is very short range, the interaction with vacuum has unlimited range because the weak vacuum is continuous, and we normally describe the van der Waals force as arising out of the fluctuations in vacuum (which are a result of the weak charge / vacuum dipolarity), and calculate it via the Casimir effect, which is an expression of a continuous vacuum. So this is a real link between aggregated structures at all levels. In a sense, this is the 'physical' side of 'process', and there should be a set of fundamental symmetries associated with this which operate at all levels.

An example of this may be the chirality or 'one-handedness' in both physics and biology, already mentioned, where many people have long sought a link. Thus, the spin term which emerges from nilpotent fermionic structure has an inherent bias towards left-handedness in its weak interactions. This is an intrinsic property of the nilpotent fermion operator, and it automatically implies a filled vacuum (or bias towards $+E$ and $+t$). The computer version of the rewrite system suggests that it is an aspect of defining the concept of 'negative' [Diaz and Rowlands, 2005]. Biology also has the same one-handed bias. Thus, amino acids can chemically exist as Dextro or Laevo (light) rotatory forms. However, biological systems only use L-forms, D-forms being toxic in many cases. (glycine is an exception being a symmetrical molecule.) In addition, it would seem that D-forms of sugars take precedence within Nature. Perhaps, this also has a mathematical origin.

The idea of 3-dimensionality

Studies of the universal rewrite system, etc., have shown that there are two ways of looking at 3-dimensionality: nonconserved and conserved. In the nonconserved version, as in ordinary space, the dimensions are indistinguishable. But the conserved case is a higher-order, more composite, concept, in which the 3 dimensions are different. This is what happens with the Dirac nilpotent. The $\mathbf{k}$, $\mathbf{i}$, $\mathbf{j}$ are different, and always in the same specific way. Always one dimension represents scale (say m), another dimensionality itself (say $\mathbf{p}$), and the remaining one the book-keeping (e.g. handedness) (say E). The weak interaction is the classic book-keeper. This is its only function. Biology may be using conserved 3-dimensionality and mapping it onto the nonconserved 3-dimensionality of space; and this is why the tetrahedron can reproduce the other Platonic solids and biological structures.

Biology, in fact, seems to require a double 3-dimensionality in the same way as physics does for quantum mechanics and the structure of particles [Rowlands, 1983, 1994, and contributions to ANPA]. Ilert, for example, has proved that the minimum correct representation of the growth of a sea-shell requires a doubling of conventional 3-D space [1992]. It seems to be required to describe development *in time*, which, in biology, is also connected with spirals and handedness. This would make sense if we need the 3-D of $\mathbf{k}$, $\mathbf{i}$ and $\mathbf{j}$ to relate space and time in $(\pm i\mathbf{k}t \pm i\mathbf{r} + 1\mathbf{j}\tau)$, or momentum and energy in its Fourier transform, $(\pm i\mathbf{k}E \pm i\mathbf{p} + 1\mathbf{j}m)$.

However, the structure of this object suggests that we need an additional pseudoscalar term, which according to the rewrite system is really the manifestation of an incomplete additional 3-D system. This provides the classic pattern of the ‘conserved 3-dimensionality’ associated with the creation of $(\pm i\mathbf{k}E \pm i\mathbf{p} + 1\mathbf{j}m)$ and $(\pm i\mathbf{k}t \pm i\mathbf{r} + 1\mathbf{j}\tau)$. In addition to the ordinary space-like 3-D of $\mathbf{p}$ or $\mathbf{r}$, there is also the 3-D ‘charge space’ or ‘angular momentum space’ of $\mathbf{k}$, $\mathbf{i}$ and $\mathbf{j}$, and the incomplete 3-D of iE and it .

From a fundamental point of view, we see that the development of increasing complexity is related to the existence of the fifth, symmetry-breaking, term in the fermionic state, or, more fundamentally, in the rewrite process as applied to natural phenomena. Though the mathematical structures go on to infinity, only 2.5 quaternion series appear to be needed for infinite replication in a ‘physical’ universe. At the same time, replicating biological systems seem to be generated in a way that conforms to similar mathematical processes, and to be constrained by the necessity of fitting into a context determined by the 3-dimensionality

of space. There appear to be similarities in that the 3-dimensionality has to be applied strictly, 2.5 times.

Application to Biology: DNA and RNA Structure

Several important aspects seem to relate fundamental physical structures to those of biology, as though the process was somehow universal and independent of level:

- The concept of double helix

- The significance of 64 units

- Chirality

- The relevance of Platonic solids (tetrahedra, cubes, etc.)

- The 5-fold broken symmetry (Fibonacci numbers)

We have already made some mention of the double helix and chirality, but all of these relationships are particularly relevant to DNA, RNA, and their action in coding to produce amino acids for proteins.

Both DNA and RNA are macromolecules and they are polymers, the monomers of which are called nucleotides. A single nucleotide consists of a 5-carbon sugar, either deoxyribose (DNA) or ribose (RNA), one or more phosphate groups and a nitrogen containing base. Four different nucleotides are found in DNA differing only in the nitrogenous base. The nucleotides are given one letter abbreviations as shorthand for the four bases; adenine (A) guanine (G) cytosine (C) and thymine (T). RNA is also comprised of 4 different bases and the same convention is followed for nomenclature, however in RNA the base thymine (T) is replaced by the base uracil and a single letter U is employed as shorthand.

Within a cell DNA is present as a supercoiled double stranded macromolecule; two polynucleotide chains held together by weak thermodynamic forces (hydrogen bonds) to form a double stranded right handed helical spiral. The two polynucleotide chains 'run' in opposite directions with A always pairing with T and G with C. For a given protein one of the two strands is referred to as the sense (coding) strand and the second the antisense (non-coding) strand. Some idea of the complexity of this system may be reached by reflecting upon the fact that the total amount of DNA in a single human cell is approximately 1.8 m long and 4nm wide.

Proteins are made up of amino acids. The precise number, and sequence, of amino acids makes up the primary structure of a polypeptide chain. A functional protein may consist of a single, or several polypeptide chains.

DNA must therefore carry genetic information that determines not only the number and types of amino acids that appear in a polypeptide, but also their exact sequence in the chain. The code for this primary structure cannot be carried within the sugar-phosphate 'backbone' of DNA since this part of the structure is identical in all DNA, variation only occurs in the base sequence. Therefore the sequence of bases in DNA determines the sequence of amino acids in a polypeptide chain. However it is apparent that to use the sequence of 4 different bases to code for 20 different amino acids the code cannot be as simple as 1 base coding for 1 amino acid. In fact each amino acid is coded for by a sequence of 3 nucleotide bases, this allows 64 variants. A triplet of bases is called a codon.

Transcription

To access the genetic information incorporated in DNA it must be converted into a usable molecular 'format.' This is the so called messenger RNA (mRNA). The process by which this is achieved is called *Transcription*. During transcription a single strand of mRNA is synthesised using a double stranded DNA molecule as a template. The two strands of the DNA molecule are separated from one another, exposing the nitrogenous bases. Only one strand is actively used as a template in the transcription process. The RNA sequence that is made is a direct copy of the nitrogenous bases in the DNA sense strand *but* it is the complementary sequence. If a guanine (G) base is part of the sequence on the sense DNA strand, then the RNA molecule has a cytosine (C) base added to its sequence at that point. In the RNA molecule uracil (U) substitutes for thymine (T). In this way the process of transcription constructs a small (relatively), mobile mRNA molecule comprised of a nucleotide sequence which is complementary to a coding sequence in the DNA molecule.

Translation and triplet codons

Translation is the process by which the genetic information in mRNA directs the synthesis of a polypeptide by controlling the order of insertion of amino acids into the growing polypeptide. This is achieved by means of a second much smaller form of RNA namely transfer RNA (tRNA). A cell contains about 60 different types of tRNA, it is mainly single stranded, 70-90 nucleotides long, and some portions of the molecule are double stranded giving the whole molecule a cloverleaf shape. Each tRNA molecule possesses two important features:

1. An anti codon site, which consists of a triplet of unpaired bases. The sequence of bases in this site varies from molecule to molecule and there is an anti codon sequence that is complementary to each codon sequence found on mRNA.

2. An amino acid binding site at the free end of the molecule that can bind a specific amino acid.

The particular amino acid that binds to each tRNA molecule is determined by the anti codon sequence. The actual amino acid is that which would be specified by the nucleotide sequence complementary to the anti codon, i.e. the codon on mRNA. For example the mRNA codon UCU specifies the amino acid serine. Thus the tRNA molecule that could recognise and bind serine would carry the anti codon AGA. Consider the whole process:- The DNA triplet codon for the amino acid serine is AGA this is transcribed into the complimentary mRNA triplet codon UCU. This codon is recognised by a tRNA molecule carrying the codon AGA and this tRNA is attached to the amino acid serine.

There are 64 (4^3) different possible triplets that can be obtained from the four DNA bases A, T, G and C which theoretically could code for 64 different protein building blocks (amino acids) but Nature (generally) selects only 20 amino acids which can be coded for by 1-4 triplets as shown in Table 1.

		2nd Position							
		U		C		A		G	
1st Position 5' end	U	UUU UUC UUA UUG	Phe Phe Leu Leu*	UCU UCC UCA UUG	Ser Ser Ser Ser	UAU UAC UAA UAG	Tyr Tyr STOP STOP	UGU UGC UGA UGG	Cys Cys STOP Trp
	C	CUU CUC CUA CUG	Leu Leu Leu Leu*	CCU CCC CCA CCG	Pro Pro Pro Pro	CAU CAC CAA CAG	His His Gln Gln	CGU CGC CGA CGG	Arg Arg Arg Arg
	A	AUU AUC AUA AUG	Ile Ile Ile* Met*	ACU ACC ACA ACG	Thr Thr Thr Thr	AAU AAC AAA AAG	Asn Asn Lys Lys	AGU AGC AGA AGG	Ser Ser Arg Arg
	G	GUU GUC GUA GUG	Val Val Val Val*	GCU GCC GCA GCG	Ala Ala Ala Ala	GAU GAC GAA GAG	Asp Asp Glu Glu	GGU GGC GGA GGG	Gly Gly Gly Gly
								3rd Position 3' end	

**Table 1. The 64 Triplets, 20 Amino Acids and Stop/Start*
Codons of The Genetic Code.**

Triplet codons and the Dirac algebra

Is there a link between the 64 amino acid triplets and the 64 units of the Dirac algebra? Are the biological structures determined by the same kind of algebra as the physical ones? In principle, the physical structure is an example of ‘conserved 3-dimensionality’. This means that it follows the pattern of a quantity such as angular momentum, with 3 non-symmetric ‘dimensions’ representing, say, magnitude, dimensionality itself, and handedness; or scale, dimensionality and ‘book-keeping’. In the Dirac nilpotent, these concepts are represented by the respective terms jm , ip and ikE . The structure always requires the interlocking of two complete quaternionic sets and an incomplete one (manifests itself as the pseudoscalar term); and it seems to occur in every case in which a 3-dimensional quantity becomes a conserved one.

Such an algebraic structure always requires 64 terms, and it is worth examining the structure of the units of the genetic code to see if they can be related to the known algebraic structure of the Dirac fermionic state. In the case of the amino acid triplets, the actual structure in physical space may be considered the conserved object, and the 3-dimensionality is that of space itself. Classifying the codons, we find that there are

$$4 = 4 \times 1 \text{ with 3 letters the same}$$

$$24 = 4 \times 3 \times 2 \text{ with 3 letters different}$$

$$36 = 4 \times 3 \times 3 \text{ with 2 letters the same}$$

The last is made up of 4 possible letters to be duplicated, 3 letters other than the first chosen, and 3 possible ways of arranging the 3 letters (that is 3 positions for the nonduplicated one). Parallel to this, the Dirac algebra has

- (A) 4 complex numbers (+ / –) (1, i)
- (B) $24 = 12 + 12$ complex 3-D objects (+ / –) (1, i) $\times$ (i, j, k)
quaternions + (+ / –) (1, i) $\times$ (i, j, k) vector (single 3-D)
- (C) 36 complex 3-D $\times$ 3-D numbers (complex vector quaternions)
(+ / –) (1, i) $\times$ (i, j, k) $\times$ (i, j, k) (double 3-D)

To apply this to the genetic structures we could take the (+ / –) (1, i) to represent the 4 options for bases. A and U or T could be say +1 and –1; and G and C + i and – i . In each case above, this is where the 4 comes from. The groups of 3, etc. would then be to do with the amount of variation allowed. They are certainly connected with dimensionality. It is interesting to see if the 20 that produce the amino acids can be written out to fit into this pattern. Any given representation of the Dirac nilpotent really only uses 20 units of the 64-part algebra, say:

ik	$\ddot{i}$	$\ddot{j}$	ik	j
k	$i\ddot{i}$	$i\ddot{j}$	iik	ij
$-k$	$-i\ddot{i}$	$-i\ddot{j}$	$-iik$	$-ij$
$-ik$	$-\ddot{i}$	$-\ddot{j}$	$-ik$	$-j$

That is, it uses none of (A)

8 of (B) $(+/-) (1, i) \times (j, k)$

12 of (C) $(+/-) (1, i) \times (i) \times (i, j, k)$

So, in a nilpotent fermionic structure such as $(\pm kE \pm i\ddot{i}p_x \pm i\ddot{j}p_y \pm i\ddot{k}p_z + ijm)$, there will be just 20 algebraic terms from the 64 that are important, if we want to represent particle and antiparticle, and also vacuum. (Here, we need to take into account multiplying by -1 and or $\pm i$.) We could guess that these match up to the amino acid-producing codons using:

8 with 3 letters different	$\pm k \pm ik$	$\pm ij \pm j$
12 with 2 letters the same	$\pm i\ddot{i}$	$\pm \ddot{i}$
	$\pm i\ddot{j}$	$\pm \ddot{j}$
	$\pm i\ddot{k}$	$\pm \ddot{k}$

We require a third of all possible codons with 3 letters different (8 out of 24), e.g. those in which a codon has its own $+/-$ partner only on the outside.

A G U	A C U
U G A	U C A
G A C	G U C
C A G	C U G

Again, we only want one third of the 36 codons with two letters the same (12 out of 36), e.g. those again with their own $+/-$ partners. Effectively, after we have excluded those with three identical letters (equivalent to $(+/-) (1, i)$), we want a symmetrical third of the remaining codons. Of course, biological molecules will not be as precise as single physical particles, so we will have variation (and perhaps ‘mimicry’). But we might expect an outline pattern of some kind to emerge which reflects this algebraic patterning, and preliminary analysis of the codons producing the amino acids in various species suggests that the 12 / 8 split has at least approximate validity.

For a codon with three identical bases, we should imagine a version of the 20 significant Dirac units as multiplied throughout by quaternion j . This still produces the same quaternionic units for the top four pairs in each of the two columns set out above, but the last two pairs would become $+/-i$ and $+/-1$, which are the four terms that correspond with the four codons with three bases the same. The strong showing of these codons is particularly interesting because it goes against absolute randomness over the whole 64 being the reason for the 12 / 8 split. The randomness, such as it is, is only between the quaternion elements. We could imagine that, at the most primitive level, the pattern is not random,

but has a highly structured pattern, though higher species might diverge increasingly from the original blueprint, and such an analysis might prove to be a way of measuring evolution. What seems most likely to be the case is that the initial life-forming process follows the mathematical rule because of the demands of conserved 3-dimensionality, and those of self-assembly related to fermionic structures, but, of course, in very complicated life forms the pattern provides only a very general constraint.

The Five Platonic Solids

At an even more fundamental level, the structure of DNA and the genetic code are related to the more basic ideas of Platonic solids and the Fibonacci sequence. A significant aspect of 3-dimensional space is that there are exactly five Platonic solids, or convex polyhedra with equivalent faces constructed of congruent convex regular polygons. These are the cube, dodecahedron, icosahedron, octahedron and tetrahedron (Fig 1.).

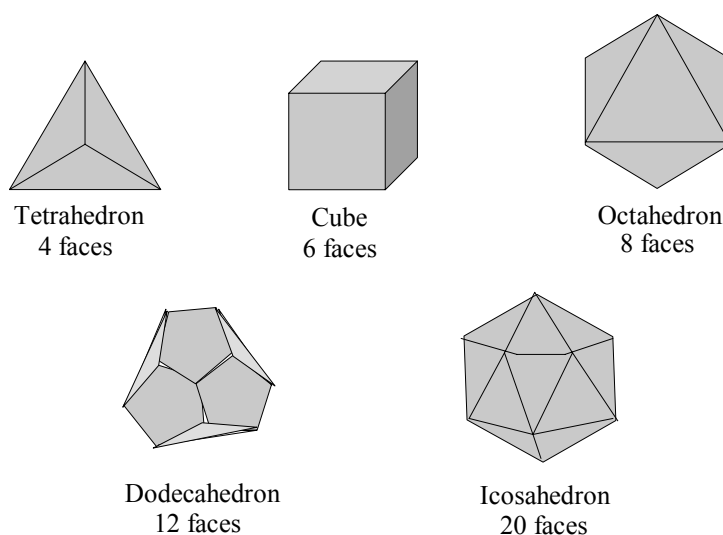


Fig 1. The 5 Platonic Solids

Significantly, the tetrahedron is a mathematical reciprocal of itself, the octahedron is a reciprocal of the cube and the dodecahedron is a reciprocal of the icosahedron and visa versa. These solids nest one within the other as shown in Fig 2.

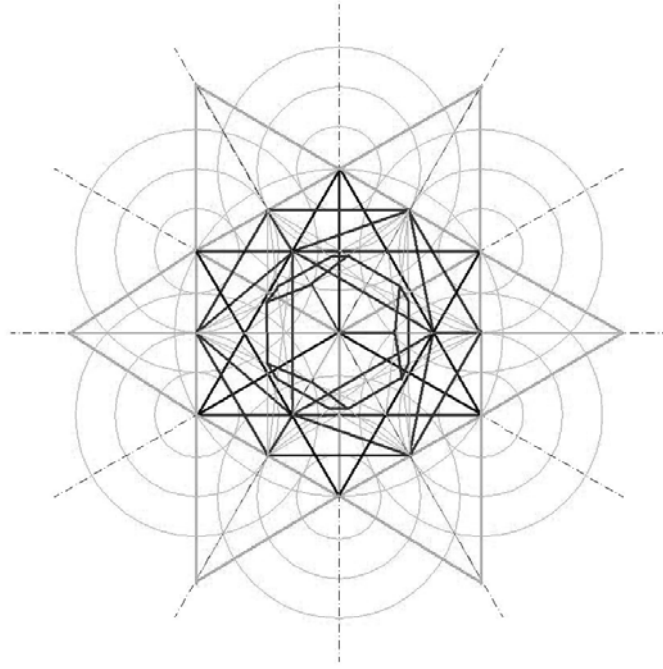


Fig 2. Nesting of The Five Platonic Solids.

The 4 fundamental parameters involved in the fermionic state, which are fundamental components of the rewrite system, can be easily represented by either a cube or tetrahedron, or, if we include the vacuum (or dual) state as well, a star tetrahedron. The rewrite system shows that 3-D forces the parameters into a 3-fold separation of dual properties / antiproperties: real / imaginary, conserved / nonconserved, discrete / continuous (= 3-dimensional / nondimensional) [Rowlands 2003a, 2004].

As the tetrahedron is the reciprocal of itself, we can also use a tetrahedral representation in which the faces / vertices represent fermion / vacuum states, and are totally reciprocal. We can also use a star tetrahedron to represent this, or relate it to the cubical structure. The cube or tetrahedron can additionally be used to represent the 8 fundamental units of the algebra required by the fermionic state, as can the octahedron, reciprocal of the cube).

In a Platonic solid, if m polygons meet at a vertex, and each polygon has n vertices, then the number of faces $f = 2 + e + v$, where the number of edges $e = f / 2$ and vertices the number of vertices $v = nf / m$, each of these being determined uniquely. (Significantly m and n can only be 3, 4 or 5 in 3-D space). These are the obvious ‘dimensionalities’ to result from the group plus dual, and the 8 primitive units of space, time, mass, charge.) The tetrahedron shows the set and the dual set, using the six edges coloured to represent the properties and antiproperties; and the vertices or faces to show *either* the set or the dual set. The next tetrahedron filling space would be opposite, like fermion / vacuum (which is what the set / dual set is really all about). If we flatten the

tetrahedral representation onto a plane we obtain the kites / darts needed for 5-fold Penrose tiling.

In general terms, the Platonic solids produce pentagonal symmetries which are intrinsic to them as a result of their structuring within a conserved 3-dimensionality ($2.5 \times 3\text{-D}$). The fermionic symmetry-breaking, which does exactly this, is clearly related to that of the geometry. And Penrose tiling is not repeatable, exactly like fermionic nilpotents. Symmetry-breaking, in both cases, is related to the creation of a 5-fold structure.

As we have seen, a significant aspect of the Dirac algebra is that it does not need 8 primitive units, and + and – signs, to generate the 64 parts. *It needs only 5 composite ones.* In effect, the most efficient structure is not the most primitive one, and is also not the most symmetrical. The $1 + 3 / 1 + 3$ symmetry that can be observed in the 8 primitive units is completely broken, when we write down the 5 composite units that most efficiently produce the 64-component algebra.

Fibonacci numbers

5-fold structures, such as Penrose tiling, also introduce the Fibonacci series and the Golden Section, which is so important in biological growth, the Sunflower being a characteristic expression of its operation. The Golden Section ratio is the one that the relative numbers of the 2 different tile types (kites / darts – flattened tetrahedral) tends towards. This occurs, in the Platonic solids, with the pentagonal symmetry of its faces. In addition, the 4-D image of a tetrahedron projected onto a 2-D plane gives a star pentagon which can be constructed within the dodecahedron. (The 20 of the dodecahedron is clearly visible in physics in the structure of the nilpotent operator which has 4 groups of 5, and contains the dualities of mass-energy / charge, fermion / vacuum, space / phase space, localised / nonlocalised, etc.)

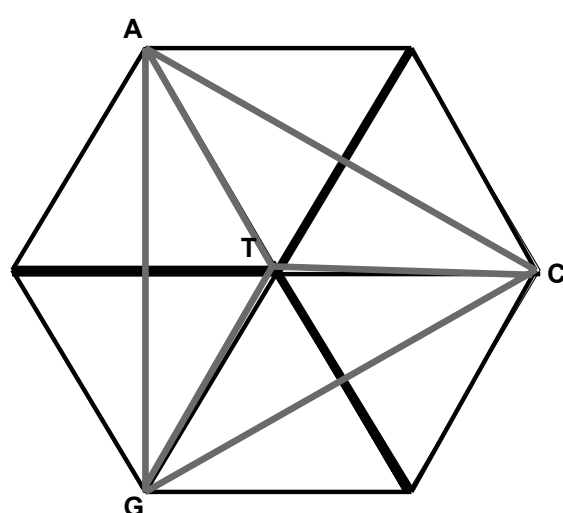
The use of the Fibonacci series as a means to explain all information processing in nature has been discussed by Stein Johansen [2005], partly in connection with the universal rewrite system. His description of the generating process as ‘1 step back and take it with you’ relates to the dual conserve / create of universal rewrite. His algorithm for information processing also splits the numbers into 5s and 3s, with a bifurcation at 8. ($8 + 3 = 11$; $8 + 5 = 13$) The 5-fold symmetry also seems to be responsible for introducing the fractal aspect in which the same patterns repeat themselves at different levels in Nature. Here, we observe that the rewrite system produces, in addition to the repetitive Clifford algebra sequence, in which the 2^n algebraic units are the basic ones, a new type of

non-repeating ‘unit of uniqueness’ which, for the first time, combines the properties of recursive and iterative systems through its symmetry-breaking 5-fold symmetric structure.

If universal rewrite is valid in the domain represented by Ilert’s work on the sea-shell, then the 5-fold Fibonacci aspect will require an additional incomplete 3-D (for time). The doubling of space is the creation of a *conserved* 3-D. If a tetrahedral arrangement is, say, the most efficient packing of 3-dimensional space, then this structure retains its identity when it is aggregated within a larger one, itself requiring 3-D spiral packing, and double helicity for stability.

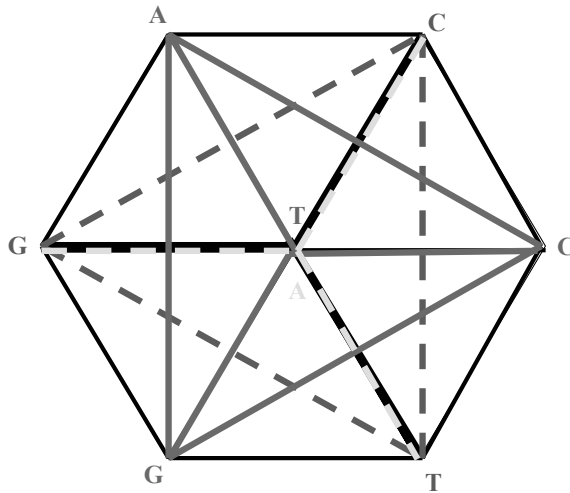
Application of geometrical structures to DNA and genetic coding

The four bases of DNA; A, T, G and C can be placed upon the four vertices of a tetrahedron (Fig 3) such that the tetrahedron can now be considered to contain, upon an information level, all the possible 64 (4^3) triplets defined by single stranded (sense) DNA or mRNA (U replacing T). Double stranded DNA can now be represented by interlocking a second tetrahedron to produce a star tetrahedron such that both the sense and antisense strands are combined with the correct base pairing of A to T and G to C that occur within the double helix (Fig 4). The corners of a cube would also serve well here.



- DNA is made up of 4 bases; A,T,G and C.
- Each base can be placed upon the corner of a Tetrahedron.
- The Tetrahedron can now be considered to contain, upon an information level, all the 64 possible triplet codons.

Fig 3. Single Stranded DNA : Tetrahedron



2 Interlocking tetrahedra :

- ‘+ sense’ strand
‘-antisense’ strand
- Note the correct ‘opposite pairing’ of A-T and G-C that exists in the double helix.

Fig 4. Double Stranded DNA : The Star Tetrahedron Within The Cube

As previously mentioned, there are 64 (4^3) different possible triplets that can be obtained from four bases. These theoretically, could code for 64 different protein building blocks (amino acids) but generally Nature selects only 20 amino acids which can be coded for by 1 to 6 different triplets as shown in Table 1. If we now look at different higher order levels of tetrahedra (Fig 5) it can be seen that the second order is composed of one octahedron and four tetrahedra and the third order is composed of four octahedra and ten tetrahedra. If each triangular octahedral face is considered to represent a single triplet then each octahedron would have eight possible triplets and if each tetrahedron is considered to represent one amino acid we would have:-

for a second order level tetrahedron : 8 triplet codons and 4 amino acids

for a third order level tetrahedron : 32 triplet codons and 10 amino acids.

Introduction of a second interlocking tetrahedral form (Fig 6) to produce a star tetrahedron would now double these values to:-

second order level star tetrahedron : 16 triplet codons and 8 amino acids

third order level star tetrahedron : 64 triplet codons and 20 amino acids.

The third order level star tetrahedron now meets the requisite numbers of triplets possible from our 4 bases and also the number of amino acids

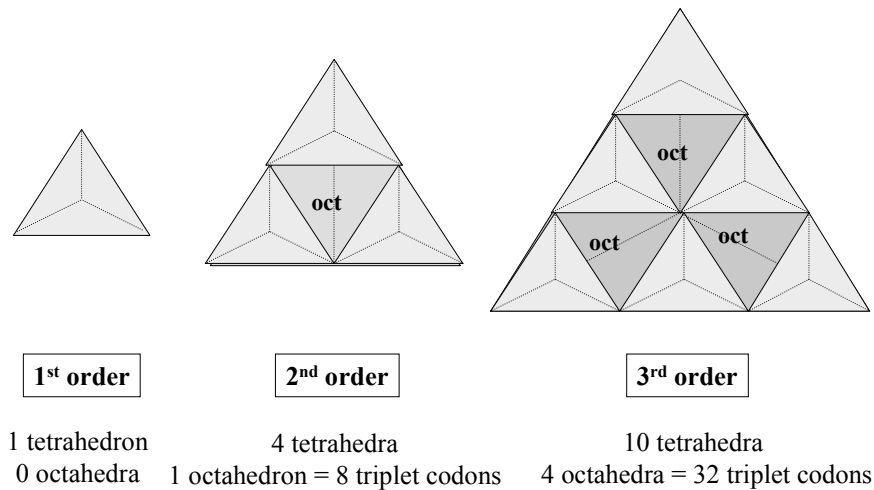


Fig 5. Higher Order Level Tetrahedra

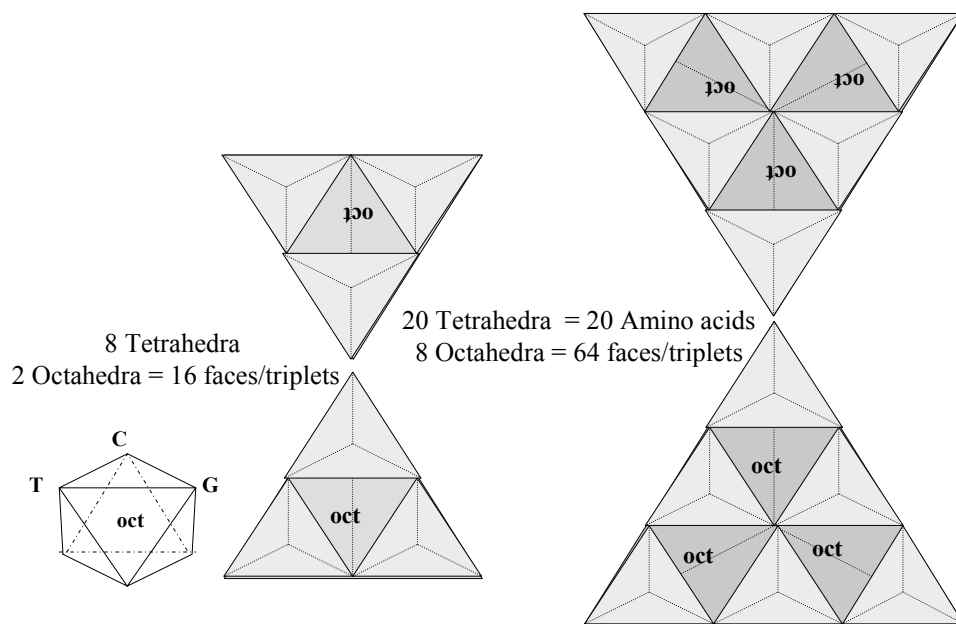
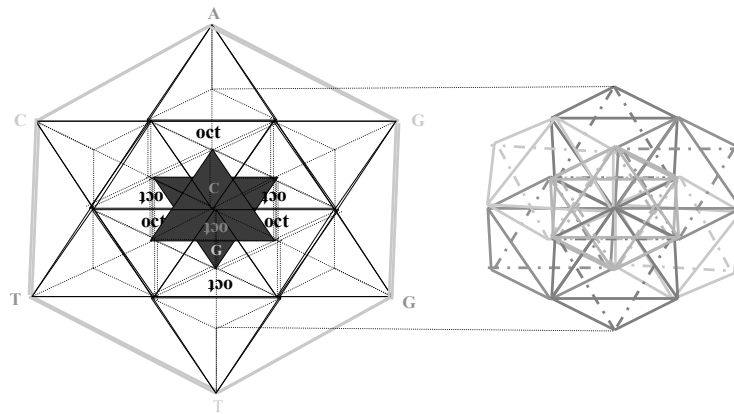


Fig 6. Star Tetrahedron : DS DNA multiply by 2

used by Nature to construct proteins. Figures 7-10 give a deeper insight into how the tetrahedra and octahedra pack within this star. Interestingly, there is a fractal nature to these diagrams, highlighting the reiteration of the star/octahedron/cube. A fractal nature is also previously observed within the higher order tetrahedra where one is reminded of the Sierpinsky's triangle.



Double stranded DNA enclosing
info of 8 octahedra
Giving $8 \times 8 = 64$ triplets and
20 tetrahedra/20 amino acids

The 8 octahedra/
64 triplet codons

Fig 7. Interlocking the Two 3rd Order Level Tetrahedra

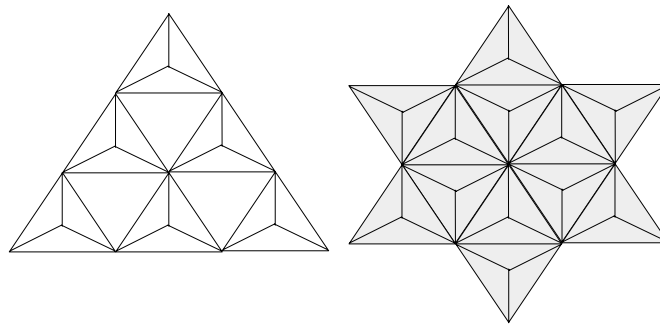
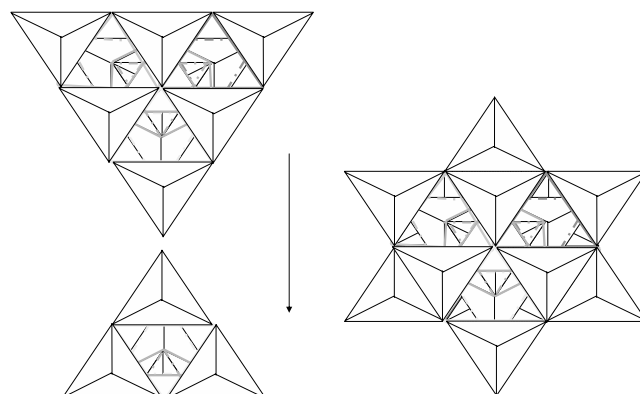


Fig 8 . Packing of the 20 Tetrahedra within the Star



**Fig 9. Completing the Star
in full 3D**

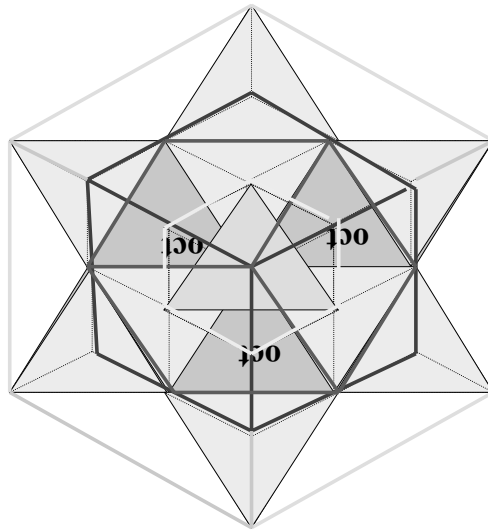


Fig 10. Cube Reiteration

The task now is to assign the correct placement of the triplets to give the appropriate amino acid represented by each tetrahedron. Figure 11 shows the beginnings of this procedure.

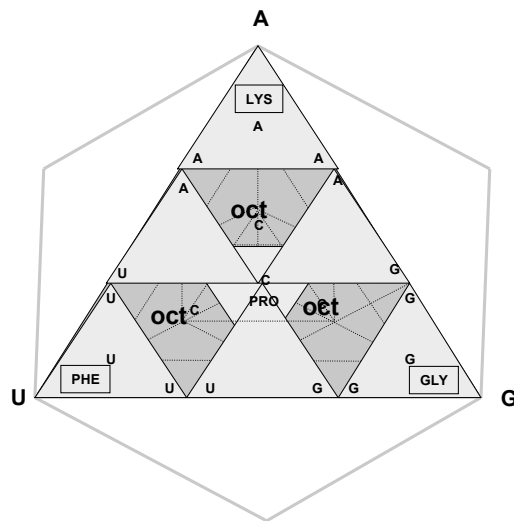


Fig 11. Beginnings of placement of the triplets and amino acids

where the corner tetrahedra can be nominated as the amino acid that relates to the triplet codon mirrored from the triangular face of an octahedron eg :-

UUU = phenylalanine (PHE)

GGG = glycine (GLY)

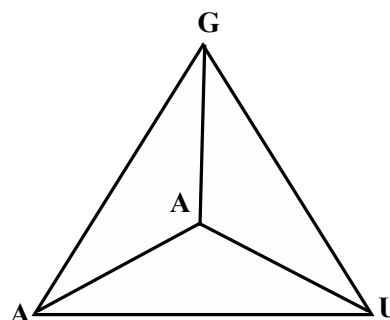
AAA = lysine (LYS)

CCC = proline (PRO)

The predominant start codon AUG, that codes for the amino acid methionine and the three common stop codons (UAG, UGA and UAA) which do not translate into any amino acids, can now all be defined by one tetrahedron as shown in Fig 12. If we consider that each face rests upon that of an octahedral ‘codon’ face then this tetrahedron is likely to be the one in the very centre of the 3rd order level tetrahedron which is completely surrounded by octahedra.

Start codon = Methionine = AUG

**Stop codons = UAG
UGA
UAA**



All 4 triplets can be defined by one tetrahedron -
perhaps the one in the very middle of the 3rd order tetrahedron

Fig 12. The Start and Stop Codons

Introducing Phi and Breaking Symmetry

In the previous section we have been mainly concerned with the 3 platonic solids of 4-fold symmetry and all these can be defined by constructing a cube with a saltire cross (x) drawn across each cuboidal face as shown in Figure 13a. Five fold symmetry is brought in by the icosahedron and it's reciprocal the dodecahedron and hence the golden section proportion Phi ($\Phi=1.618$) comes into play.

Nature expresses herself using the golden section Phi and 5 now becomes important. Humans find beauty in both hearing and seeing this relationship and for this reason Phi was used in sacred wall paintings in ancient Egypt and vigorously applied in Renaissance art. The first 3 platonic solids do not reveal this relationship but the icosahedron and dodecahedron do. Whereas the octahedron is transected by 3 squares the icosahedron has 3 golden rectangles in each of the 3 planes and the dodecahedron interestingly reveals another level of order in that in reality this shape is constructed of 5 interlocking cubes. The icosahedron is

often used by viruses and bacteriophages as it gives the greatest volume with stability eg the polio virus, hepatitis A virus and T phages. Phi is very prevalent within Nature and numerous examples of its use exist but the plant that displays this proportion in every way possible is the sunflower.

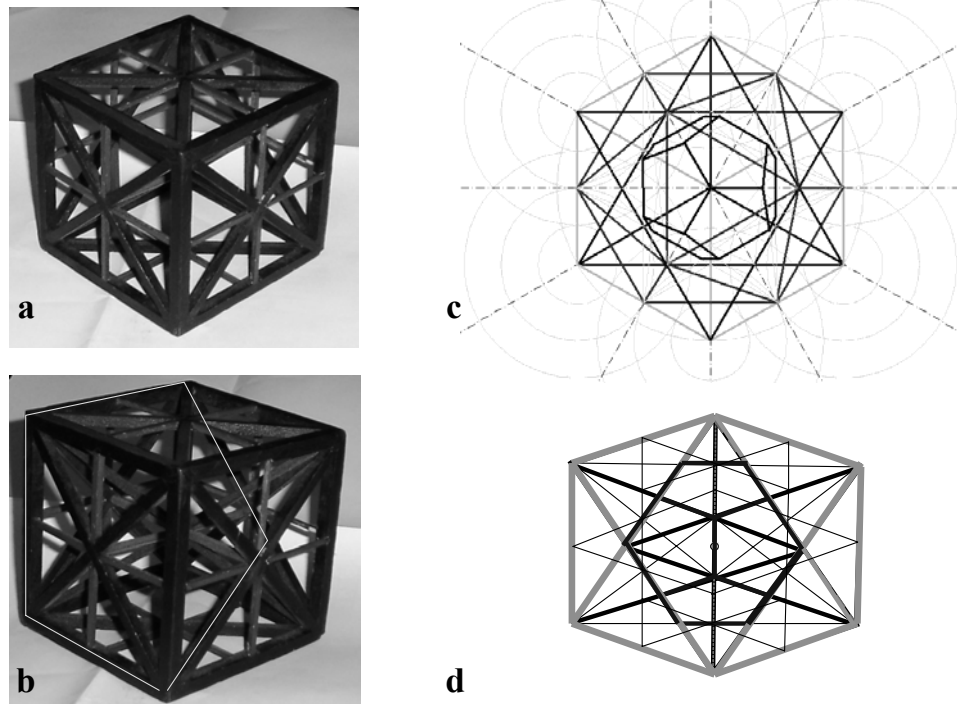
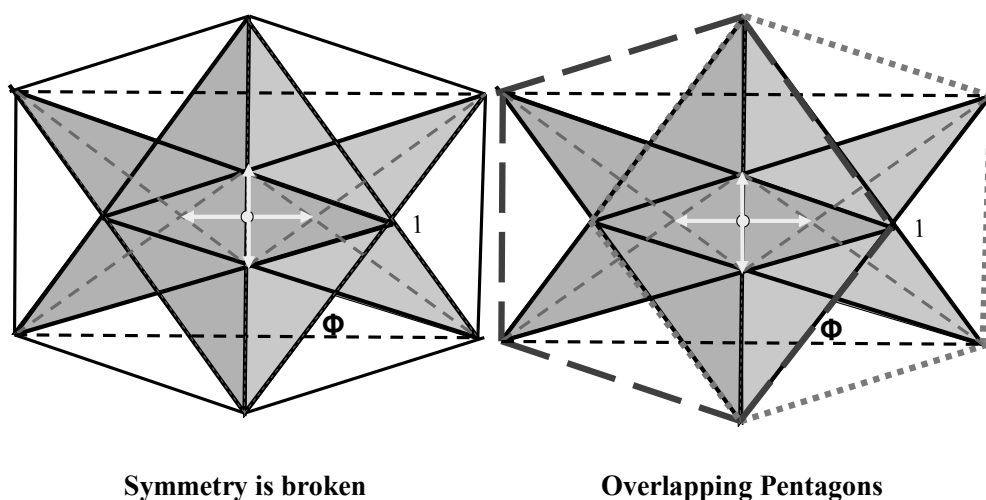


Fig 13. The Cube : Introducing Phi by Perspective Change.

Phi can be seen within our cube by drawing the second set of cross (+) lines on each cuboidal face and allowing a change in the cube's orientation such that the front corner appears to sit over a newly constructed cross point of two lines as shown in Fig 13b. This can be difficult to visualise but Fig 13c is a 2D representation of the cube in the first orientation and Fig 13d and Fig 14 shows the 2D outline of the cube in the second perspective. The latter now also gives a 2D outline of the icosahedron using exactly the same outline of this new perspective of the cube.



**Fig 14. Perspective 2 : Showing Pentagons Within The Cube-
Polarity is Introduced and Φ**

This change in cuboidal perspective appears to introduce polarity where the central point of the first perspective now opens up from a single point and expands north-south and east-west into a plane. Two overlapping pentagons with inscribed 5 pointed stars can now be defined (Fig14). This whole figure is governed by Phi and this image is also a correct 2D 'seethrough' image of an icosahedron with its reciprocal dodecahedron inside (Fig15).

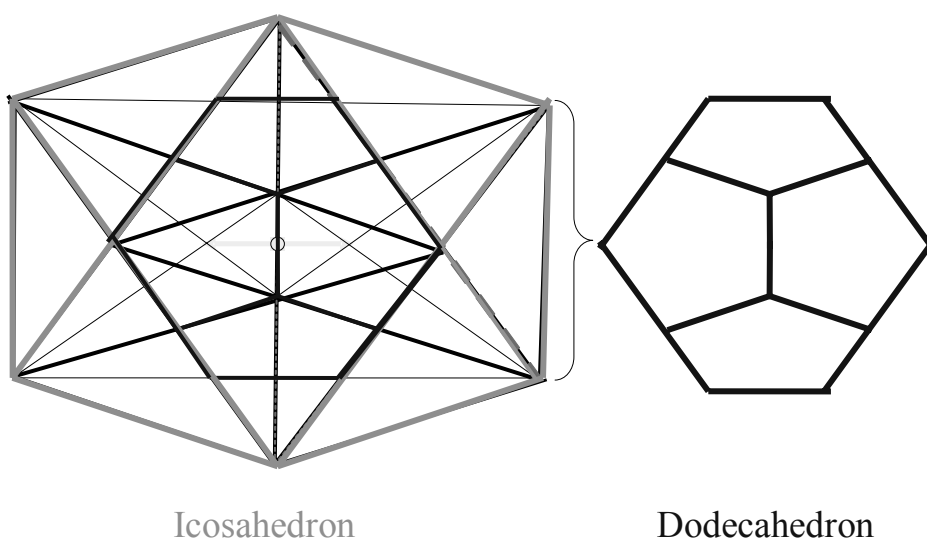


Fig 15. Icosahedron with Internal Reciprocal Dodecahedron

Within the algebra of symmetry breaking there is the reduction from the 8 to the 5-fold symmetry which we can represent by packing 5 tetrahedra into a disk (Fig 16). There is a discrepancy here of $7^{\circ} 21'$ but this may also have relevance as will be described later on.

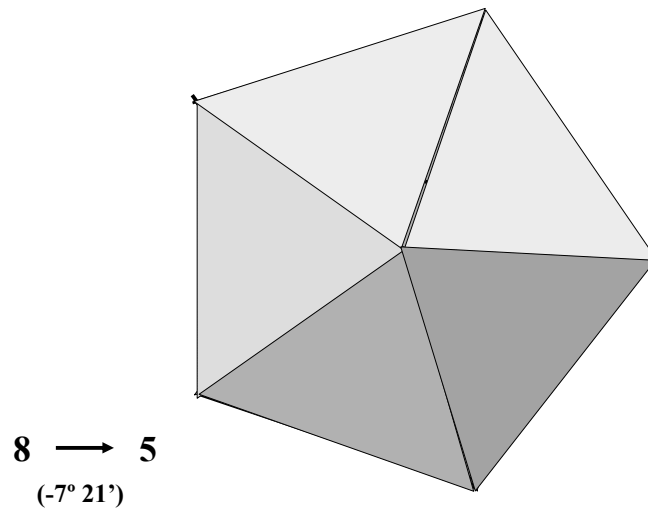
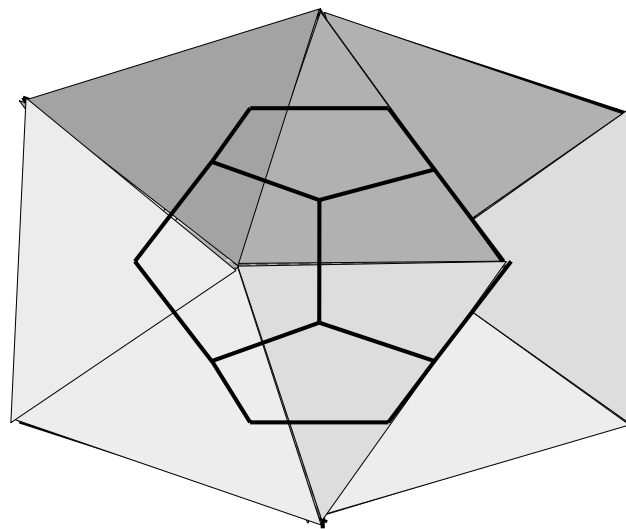


Fig 16. Breaking symmetry 5 Tetrahedrons make one pentagonal disk



2 Pentagonal discs overlaid to produce 2D icosahedron plus internal dodecahedron.

Fig 17. Icosahedron Plus Reciprocal Dodecahedron Inside

If we now overlay two pentagonal disks a 3D representation of the icosahedron can again be visualised in 2D as shown in Fig 17. Here the reciprocal internal dodecahedron has been highlighted which as we shall see has certain interesting consequences.

Fig 1 Crick and Watson pairings

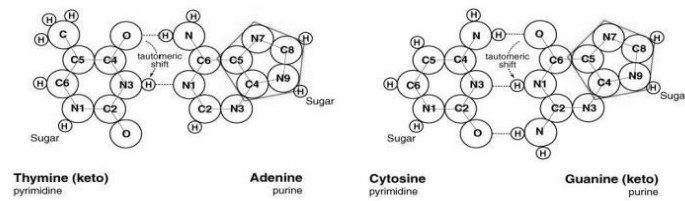


Fig 2 Proposed pentagonal pairings

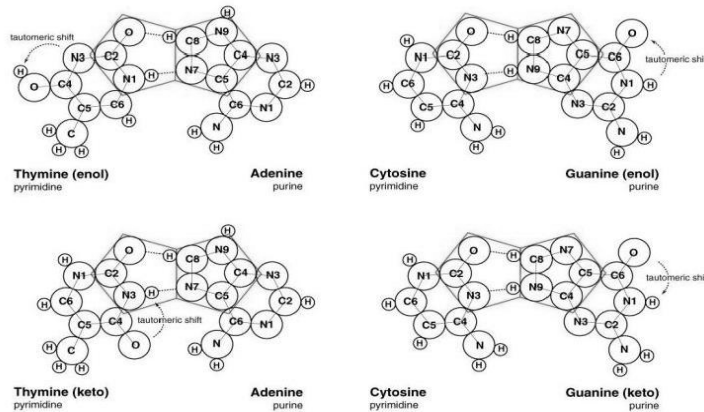


Fig 18. Mark Curtis : New structure gives exactly same DNA helix dimensions as Watson and Crick

The Watson and Crick model for DNA structure [Watson and Crick 1953] is well known but in May 1998 a second possible model was proposed by the artist Mark Curtis who designed one of the stamps for the millennium. Curtis wanted an in depth understanding of this structure and realised that something was amiss when he attempted to build models according to the Watson and Crick theory. After studying the

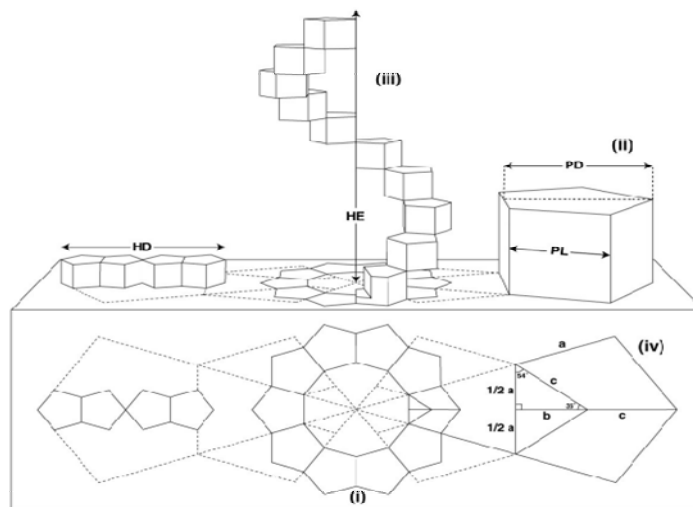


Fig 19. Curtis DNA model 1998.

original X-ray crystallography data and conferring with a chemist, it became apparent that there was a different way to pair the nucleotides within the double stranded DNA that would give the same data. The Curtis model allows for the inclusion of both possible forms of the dNTPs namely the keto and enol molecular forms of thymine and guanine whereas the Watson and Crick model only allowed for one, the keto form (Fig 18). Fig 19 shows Curtis' resultant helix as a stack of 10 pentagonal blocks representing the 10 nucleotides predicted by the X-ray crystallography data and below a view looking down the spiral of a ring of 10 pentagonal blocks/nucleotides. The stamp design Curtis finally produced was available to the public in 1998 (Fig 20).

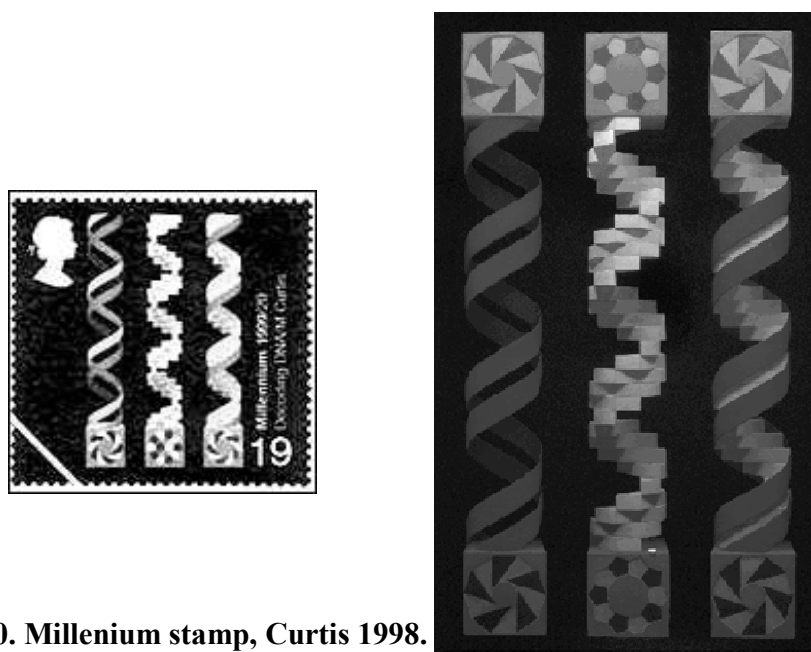


Fig 20. Millenium stamp, Curtis 1998.

It can be seen that by overlaying the icosahedral 2D outlines or the pentagonal disks over Curtis' drawings of the paired nucleotides, two of the pentagonal faces of the internal dodecahedron align with these paired nucleotides (Fig 21). If we continue this procedure further we build the same ring of pentagons displayed by Curtis for viewing the double stranded DNA spiral head on (Fig 22). The pentagons of the inner dodecahedron in Fig 22 have been highlighted with stars and can be seen to match the Curtis' ring which he represented by pentagonal blocks.

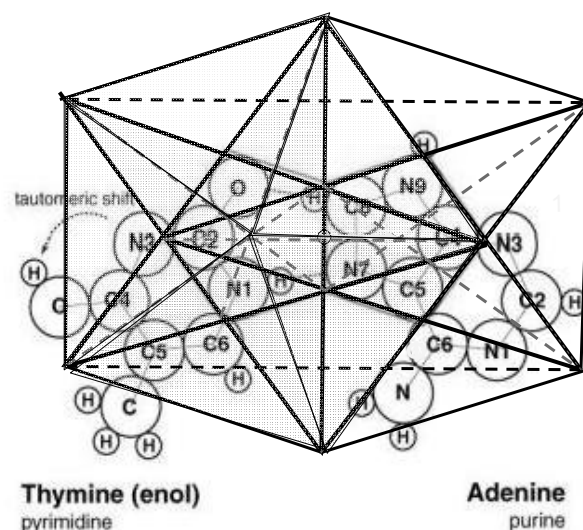


Fig 21. Enlargement of Curtis DNA model overlaid with cube (perspective 2) and pentagonal disk.

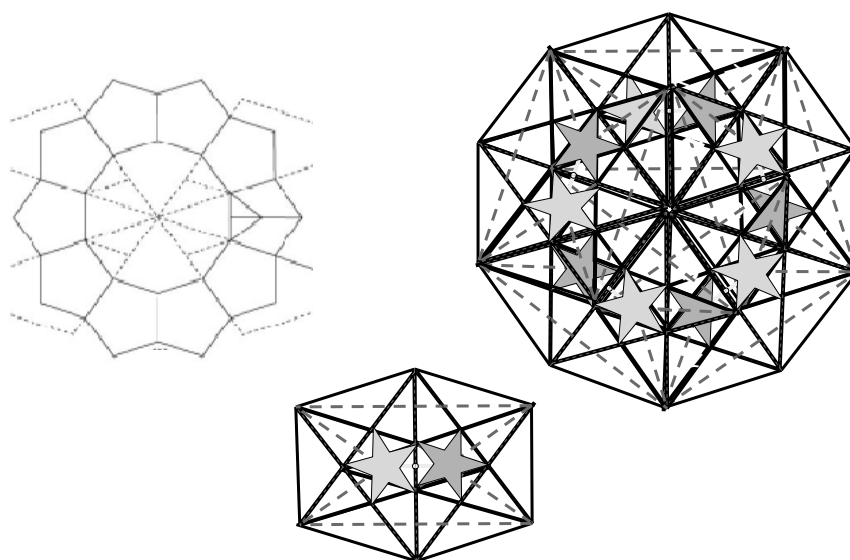


Fig 22. Adding the Icosahedra with Internal Dodecahedron To the Curtis Model.

We can of course overlay single pentagonal disks (of 5 tetrahedra) to give the same result (Fig 23). It is interesting to note that one pentagonal disk now contains 3 nucleotides of one strand of DNA. This could represent an analogy to the triplet codons of 3 nucleotides that are then translated into one amino acid protein building block when 'in frame' ie the information of the 3 is contained within the pentagonal disk of 5

tetrahedra. This is reminiscent of the algebra where the 8 is resolved into 5 when the 3 is ‘overlaid onto 5’, i.e.:

time	space	mass	charge
pseudoscalar	vector	scalar	quaternion
i	$i \ j \ k$	1	$i \ j \ k$

being reduced from eight simple to five combined or composite units, i.e.
:

ik	$ii \ ji \ ki$	$1j$
------	----------------	------

Considering the original eight, the octahedron would serve as a good candidate for this analogy. The octahedron is also the reciprocal of the cube which is here the 2D outline of our icosahedron.

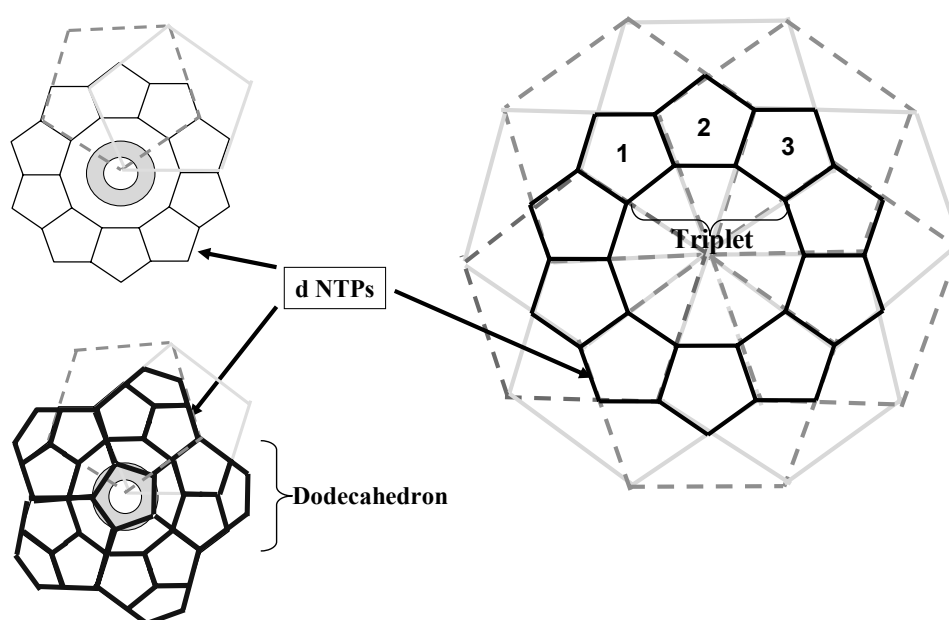


Fig 23. Highlighting The Nucleotides –One Triplet per Pentagon

The stacking of the pentagonal disks representing the DNA helix is shown in Figs 23, 24 and 26. Figs 25 and 26 are pictures of models of these diagrams made from tetrahedral dice. The model building process clarifies that 2 spirals of 5 stacked pentagonal disks carry the 10 nucleotides and that 2 strands of these fit together to produce the double stranded helix. It is interesting to note that there is a groove running around the molecule that implies space for a third strand of disks, could this be for the RNA or the stabilising water molecules? Double stranded DNA is known to unzip to allow a copy of RNA to be made which is then sent elsewhere to be translated into proteins. After the RNA copy has been completed it needs to unzip itself to release the RNA and the DNA

strands are then rezippped together. Perhaps the missing $7^{\circ} 21'$ causes the necessary instability for this zip/unzip process by giving a spring effect.

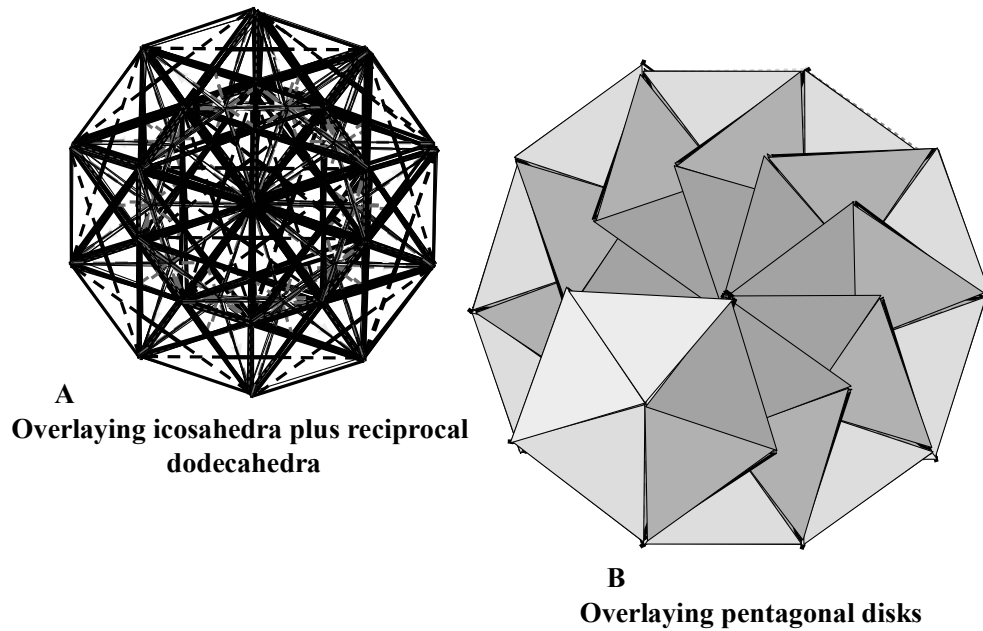


Fig 24. The DNA Helix.

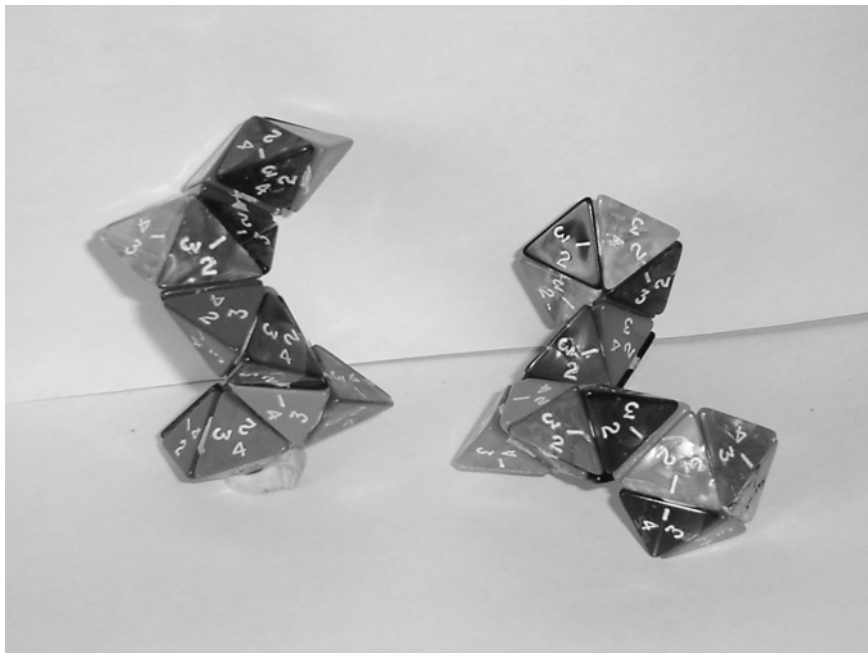


Fig 25. Two spiral strands of pentagonal disks : 5 per turn

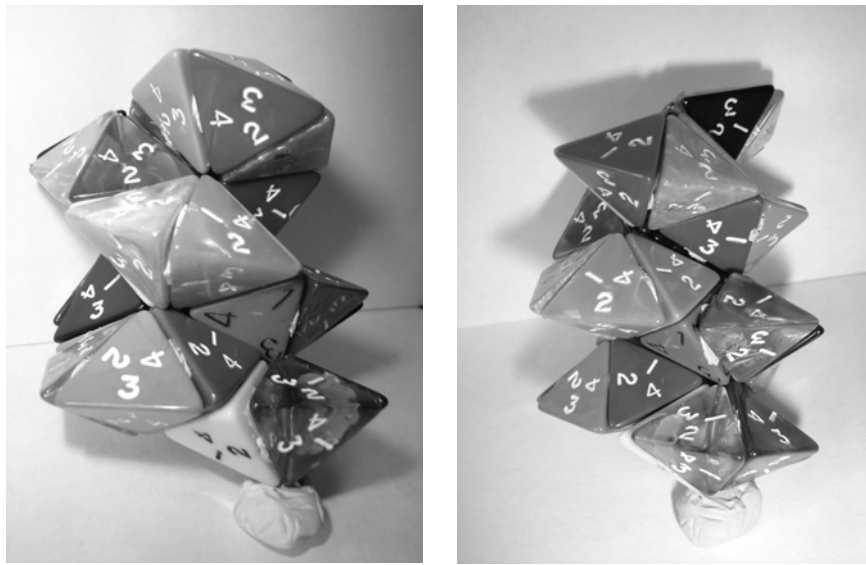


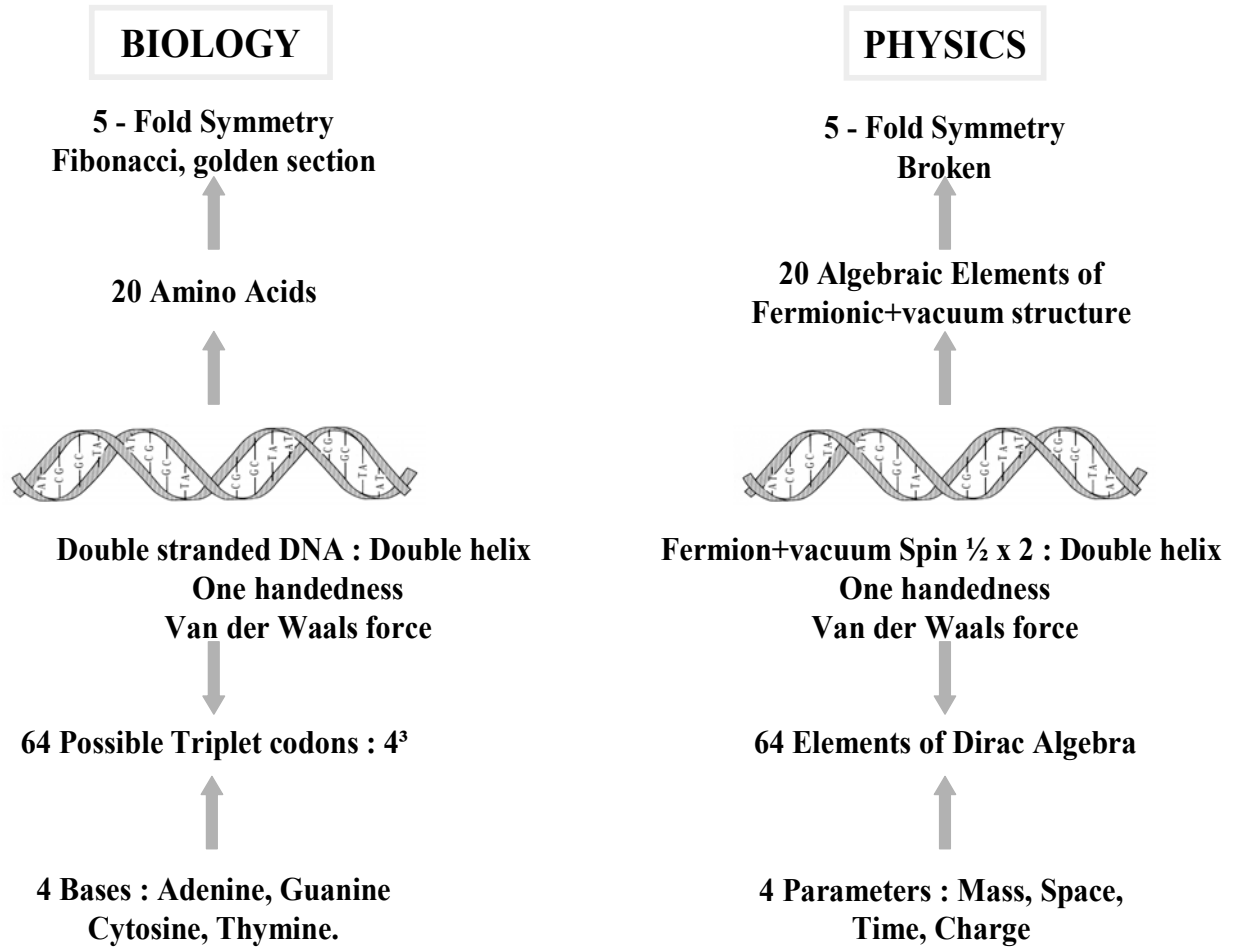
Fig 26. Double stranded DNA : 2 Spiral strands of pentagonal disks with a total of 10 disks per turn.

The need for 2×5 dNTPs for every twist of DNA is interesting. Each twist of a single strand contains 1 codon and 2 other dNTPs (cf $1 \times 3\text{-D} + 2$ others in the nilpotent). You can only guarantee a codon if there are 5 dNTPs. Now, if we start with a codon in the twist, it takes 3 sets of 5 dNTPs, i.e. 3 twists, to complete a 'cycle' and start with a codon again. Here, we may think of the Fibonacci series in the way that Johansen does. Also, the twisting in DNA is continuous – it doesn't begin and end anywhere – which is reflected in the fact that the coded part of the sequence is entirely coded, with no gaps.

Conclusion

We have explored several significant examples in which the mathematics of physics and biology seem to show similar underlying structures relating to more fundamental processes, and to the idea of 'process' itself. In addition to this, we have shown that complexity arises from simplicity in all aspects of nature because the fundamental units of nature (fermions) are symmetrical only when taken in conjunction with the rest of the universe (vacuum). All types of aggregation in matter, at all levels, physical chemical, biological, and all phase transitions, are concerned with an attempt to overcome this asymmetry. The force involved in this process is the weak force (the fifth or asymmetric term) and it is ultimately through this force that we will find a deep link between physics and biology.

Fig 27. UNIVERSAL RE-WRITE SYSTEM
UNIFICATION OF PHYSICS AND BIOLOGY



REFERENCES

- Diaz, B. and Rowlands, P. 2004, A computational path to the nilpotent Dirac equation, *International Journal of Computing Anticipatory Systems*, 16, 203-18.
- Diaz, B. and Rowlands, P. 2005, The infinite square roots of -1 , presented at CASYS, 2005, to be published in *International Journal of Computing Anticipatory Systems*.
- Ilert, C. 1992, Foundations of Theoretical Conchology ... from Self-Similarity in Non-Conservative Mechanics, Hadronic Press.
- Johansen, S. 2005, *Initiation of Hadronic Philosophy, The Philosophy underlying Hadronic Mechanics and Chemistry*, presented at workshop on hadronic mechanics, Karlstad, Sweden, 22 June 2005.
- Rowlands, P. 1983, *Speculat. Sci. Tech.*, 6, 69-80.
- Rowlands, P. 1994, *Speculat. Sci. Tech.*, 17, 279-282.

- Rowlands, P. 2003a, Representations of a Fundamental Theory, in K. Bowden (ed.), *Boundaries* (May 2003, *Proceedings of XXIV ANPA Conference*, Cambridge, August 2002), 18-47.
- Rowlands, P. 2003, The nilpotent Dirac equation and its applications in particle physics, arXiv:quant-ph/0301071.
- Rowlands, P. 2004, The origin and meaning of 3-dimensionality, K. Bowden (ed.), *Spin* (May 2004, *Proceedings of XXIII ANPA Conference*, Cambridge, August 2003), 292-323.
- Rowlands, P. and Diaz, B. 2002, A universal alphabet and rewrite system, arXiv:cs.OH/0209026.
- Watson, J. D. and Crick, F.H.C. 1953, *Nature*, 171, 737-738.
- Curtis, M. and Ochert, A. 1998, *New Scientist*, 16 May, 32-35.